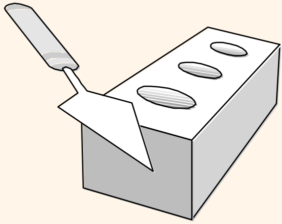
The Entity-Relationship Model

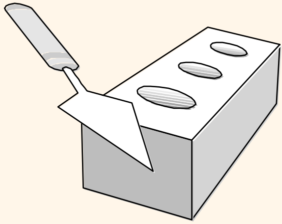
Chapter 2

Database Design Process



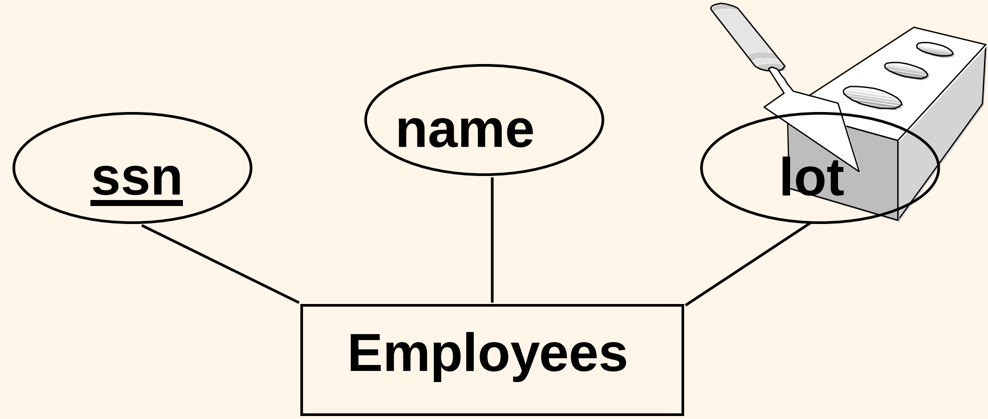
- *Requirement collection and analysis*
  - DB requirements and functional requirements
- *Conceptual DB design using a high-level model*
  - Easier to understand and communicate with others
- *Logical DB design (data model mapping)*
  - Conceptual schema is transformed from a high-level data model into implementation data model
- *Physical DB design*
  - Internal data structures and file organizations for DB are specified

Overview of Database Design



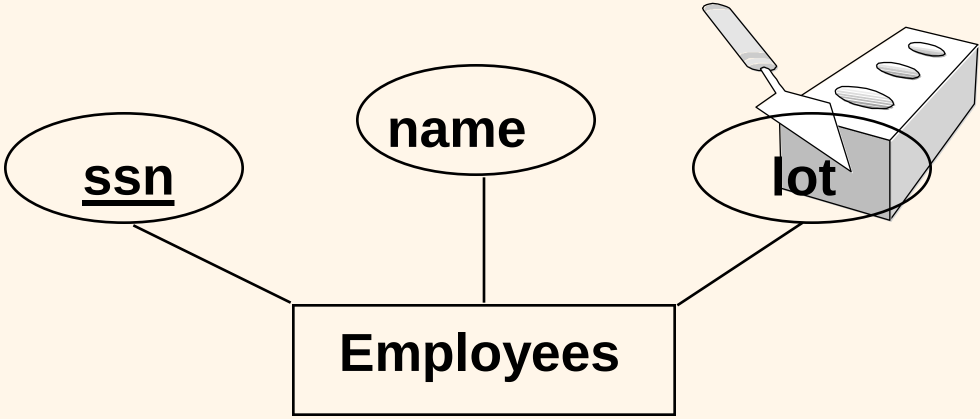
- *Conceptual design*: (*ER Model is used at this stage.)*
  - What are the *entities* and *relationships* in the enterprise?
  - What information about these entities and relationships should we store in the database?
  - What are the *integrity constraints* or *business rules* that hold?
  - A database `schema' in the ER Model can be represented pictorially (*ER diagrams*).
  - An ER diagram can be mapped into a relational schema.

ER Model Basics



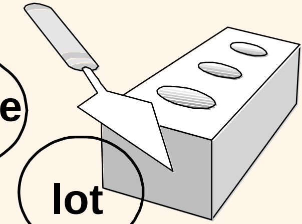
- *Entity:* Real-world object distinguishable from other objects. An entity is described (in DB) using a set of *attributes*.
- *Entity Set*: A collection of similar entities. E.g., all employees.
  - All entities in an entity set have the same set of attributes. (Until we consider ISA hierarchies, anyway!)
  - Each entity set has a *key*.
  - Each attribute has a *domain*.

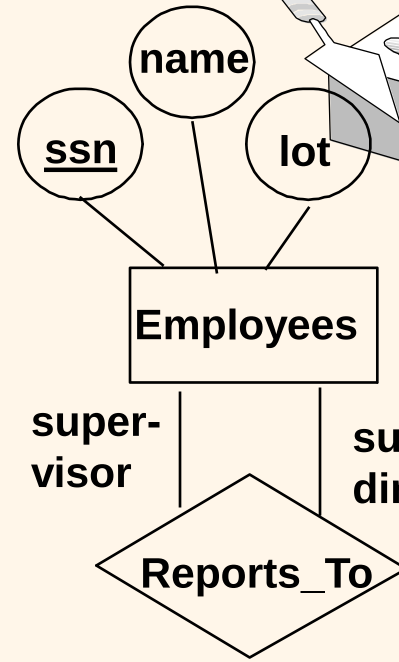
ER Model Basics

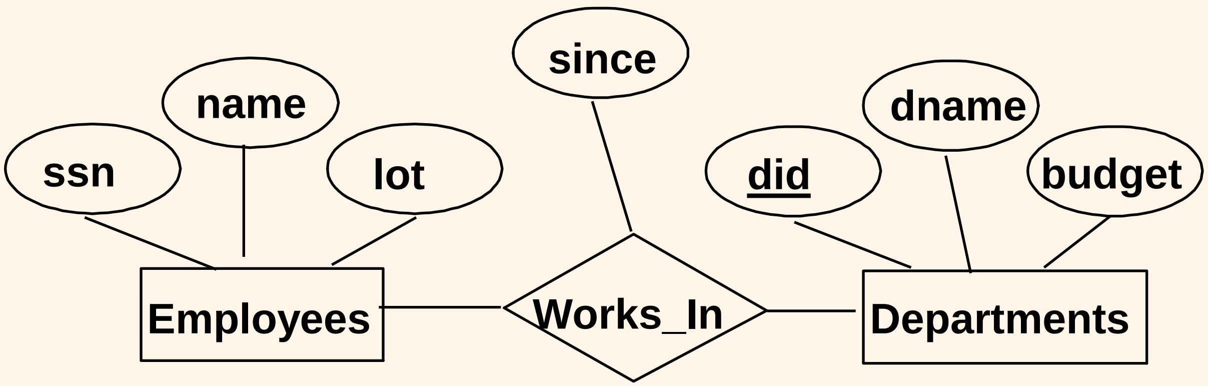


- ***Key and key attributes:***
  - Key: a unique value for an entity
  - Key attributes: a group of one or more attributes that uniquely identify an entity in the entity set
- ***Super key, candidate key, and primary key***
  - Super key: a set of attributes that allows to identify and entity uniquely in the entity set
  - Candidate key: minimal super key
    - There can be many candidate keys
  - Primary key: a candidate key chosen by the designer
    - Denoted by underlining in ER attributes

ER Model Basics (Contd.)

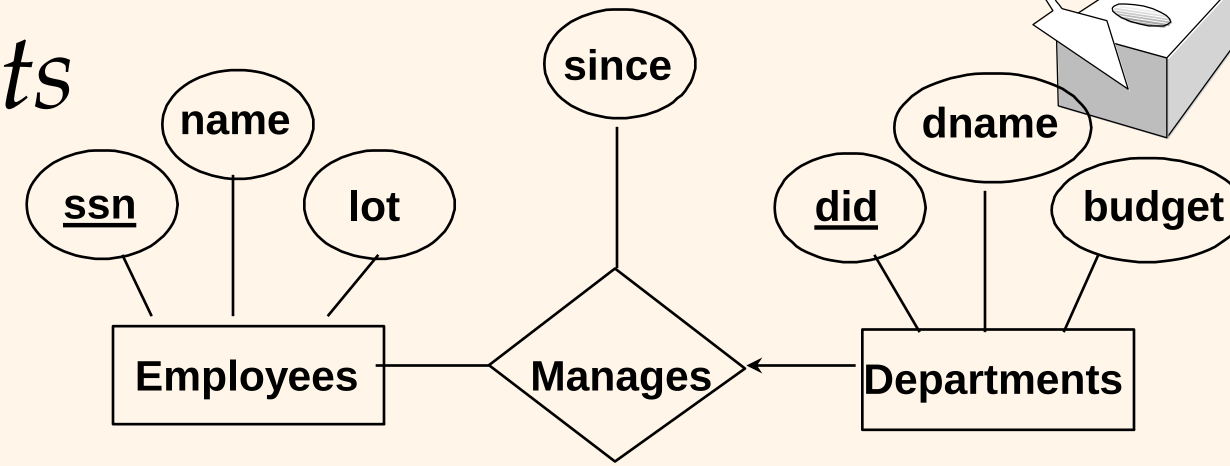




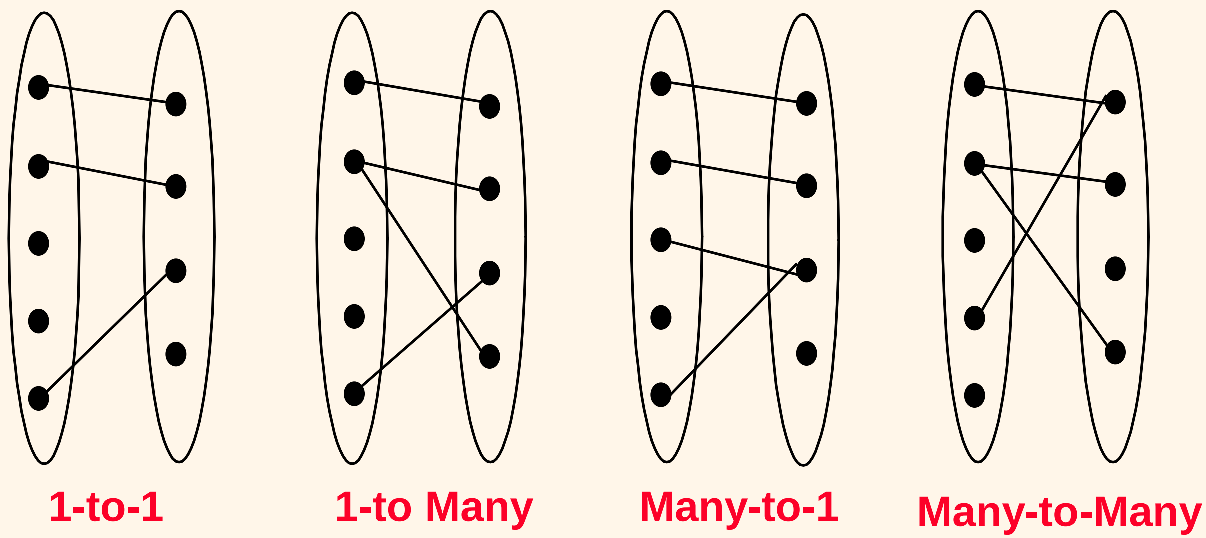


- *Relationship*: Association among two or more entities. e.g., Jack works in Pharmacy department.
- *Relationship Set*: Collection of similar relationships.
  - An n-ary relationship set R relates n entity sets E1 ... En; each relationship in R involves entities e1 in E1, ..., en in En
    - Same entity set could participate in different relationship sets, or in different "roles" in same set.

Key Constraints

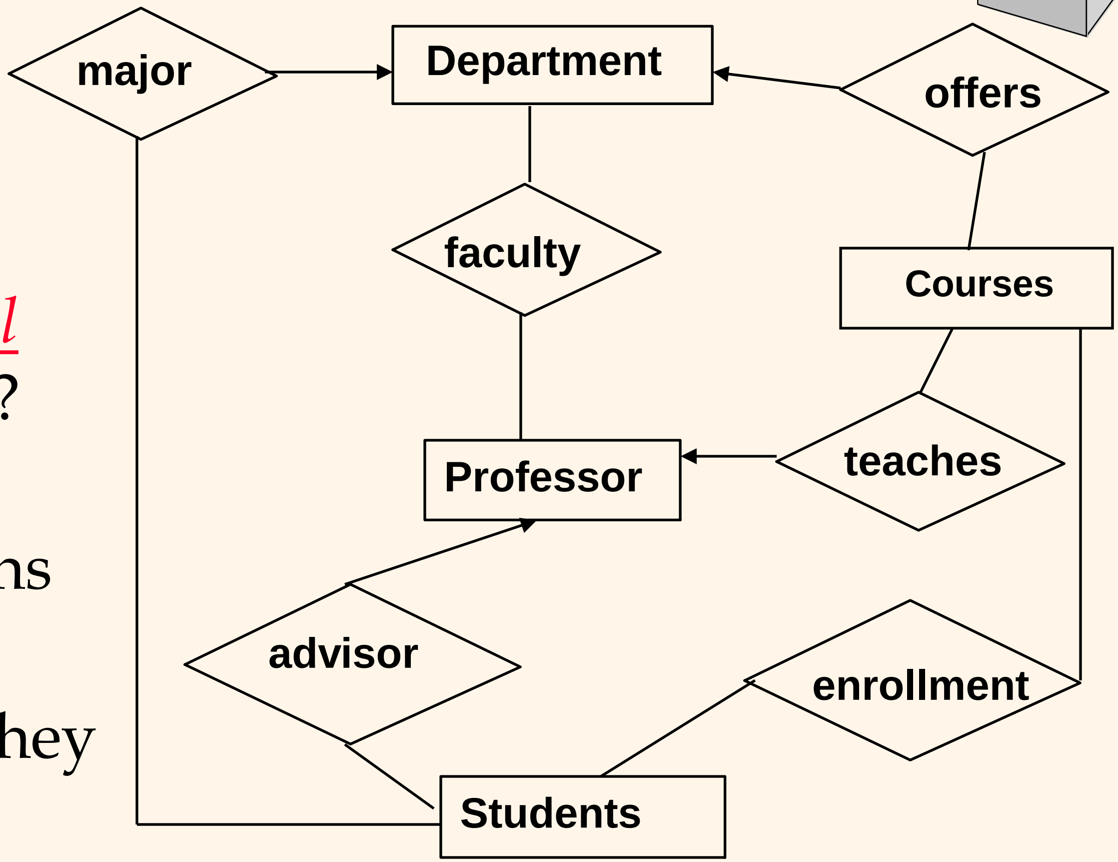


- Consider Works_In: An employee can work in many departments; a dept can have many employees.
- In contrast, each dept has at most one manager, according to the *key constraint* on Manages.



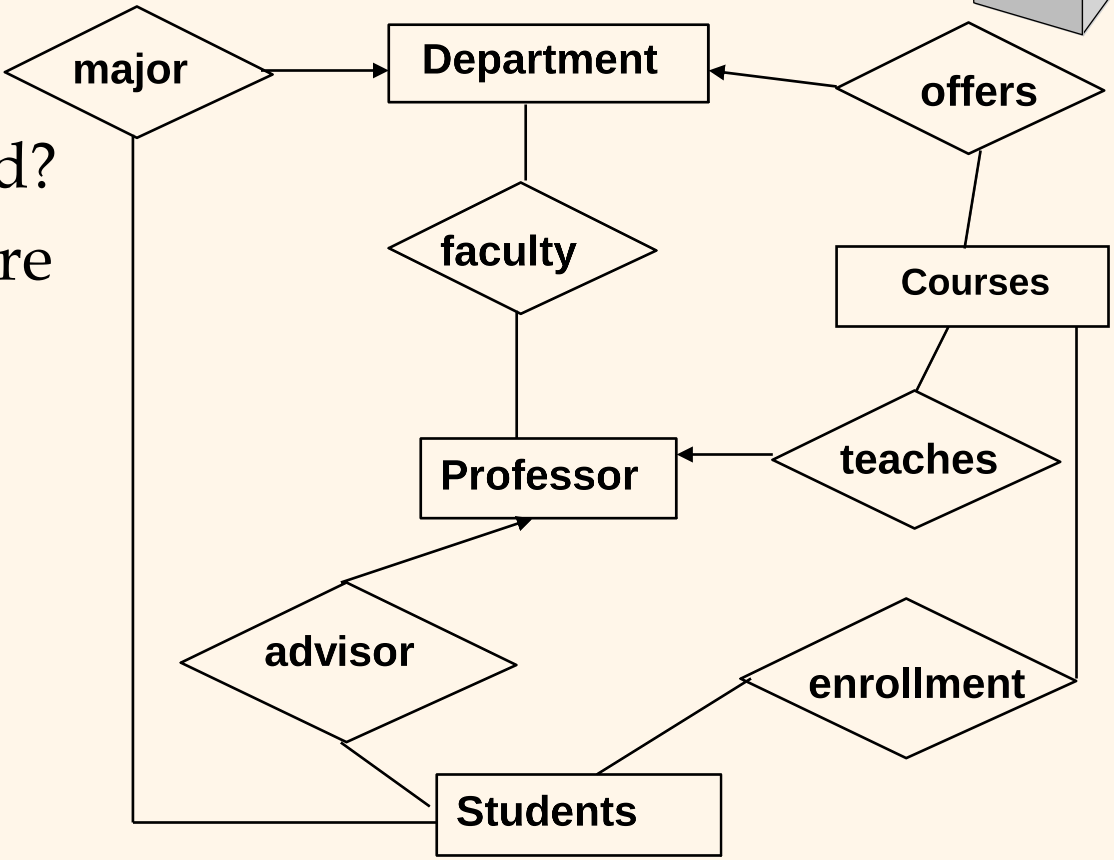
Example ER

- An ER diagram represents several *assertions about the real world*. What are they?
- When attributes are added, more assertions are made.
- How can we ensure they are correct?
- A DB is judged correct if it captures ER diagram correctly.



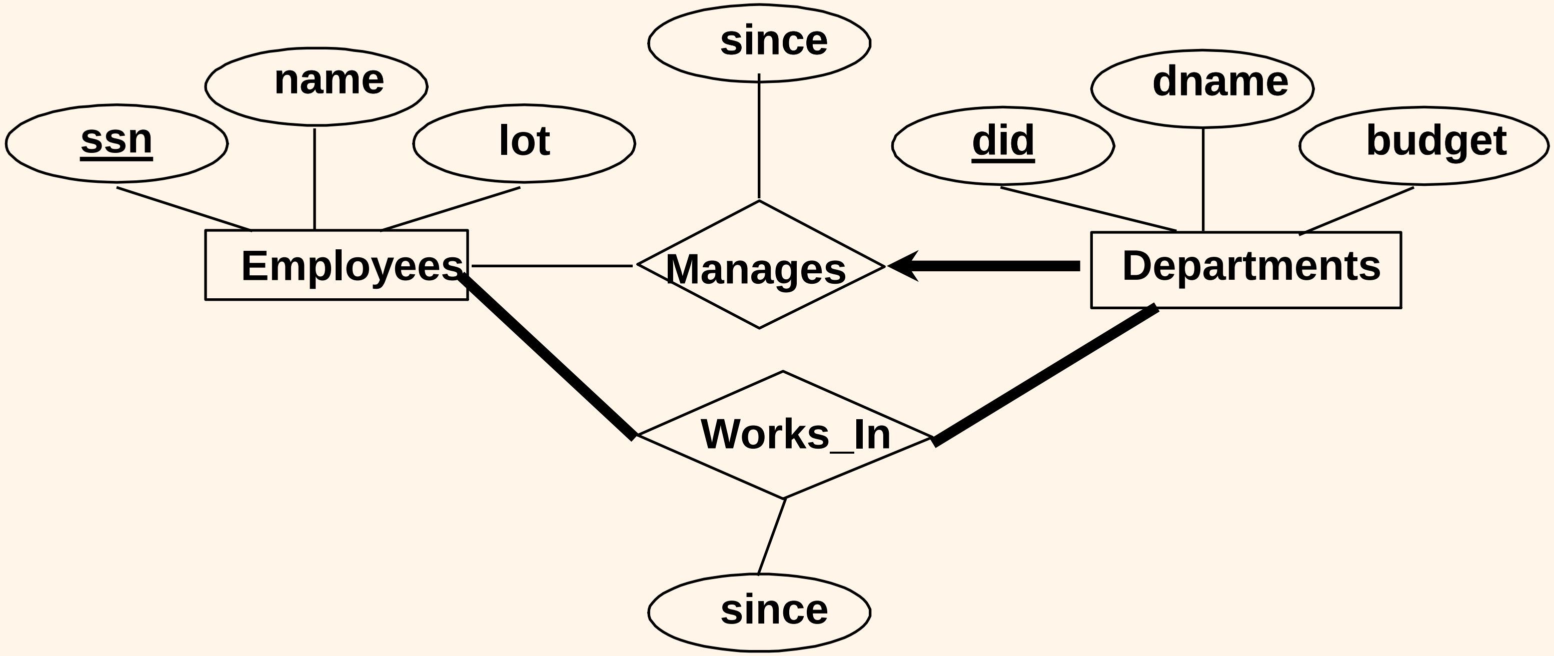
Exercises

- Is double major allowed?
- Can a student have more than 1 advisor?
- Is joint appointment of faculty possible?
- Can two profs share to teach the same course?
- Can a professor teach more than one course?
- Can a professor stay without affiliated with a department?



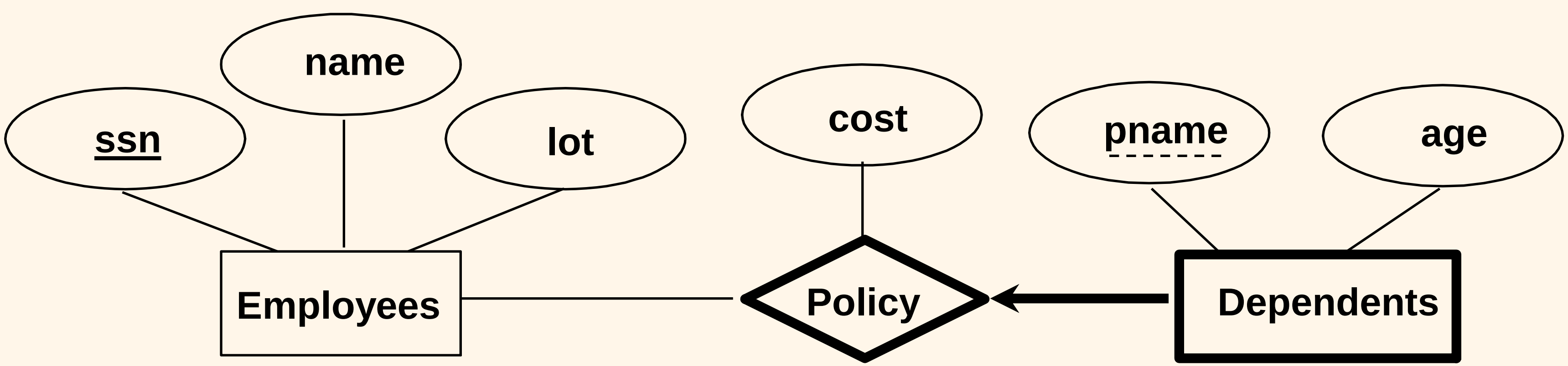
Participation Constraints

- Does every department have a manager?
  - If so, this is a *participation constraint*: the participation of Departments in Manages is said to be *total* (vs. *partial*).
    - Every Departments entity must appear in an instance of the Manages relationship.

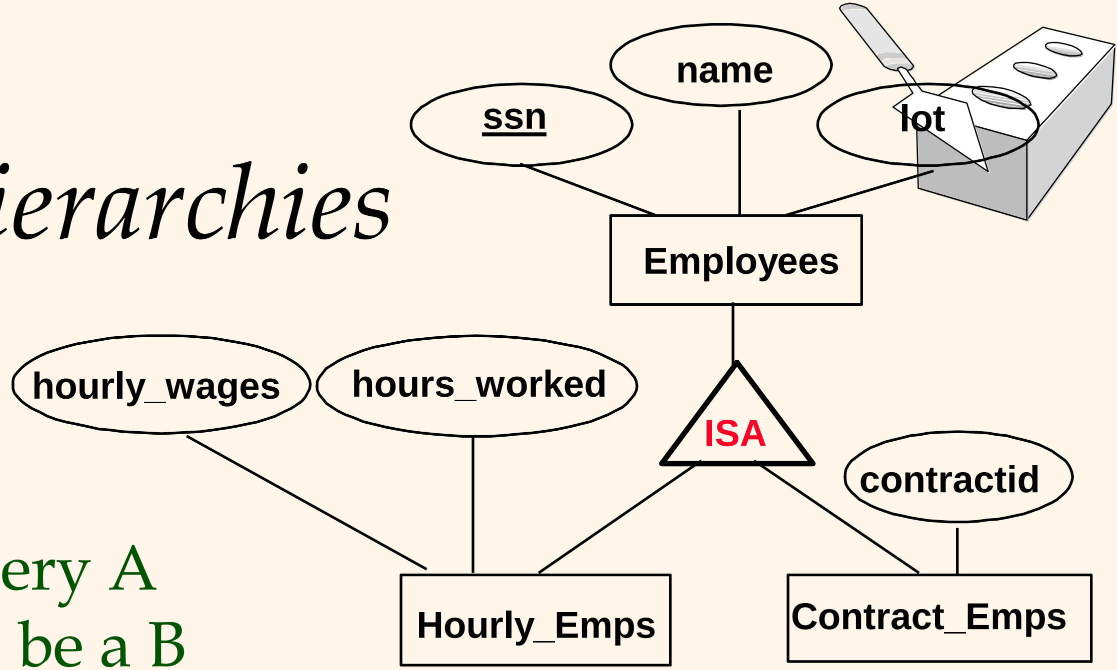


Weak Entities

- A *weak entity* can be identified uniquely only by considering the primary key of another (*owner*) entity.
  - Owner entity set and weak entity set must participate in a one-to-many relationship set (one owner, many weak entities).
  - Weak entity set must have total participation in this *identifying* relationship set.



ISA (`is a') Hierarchies



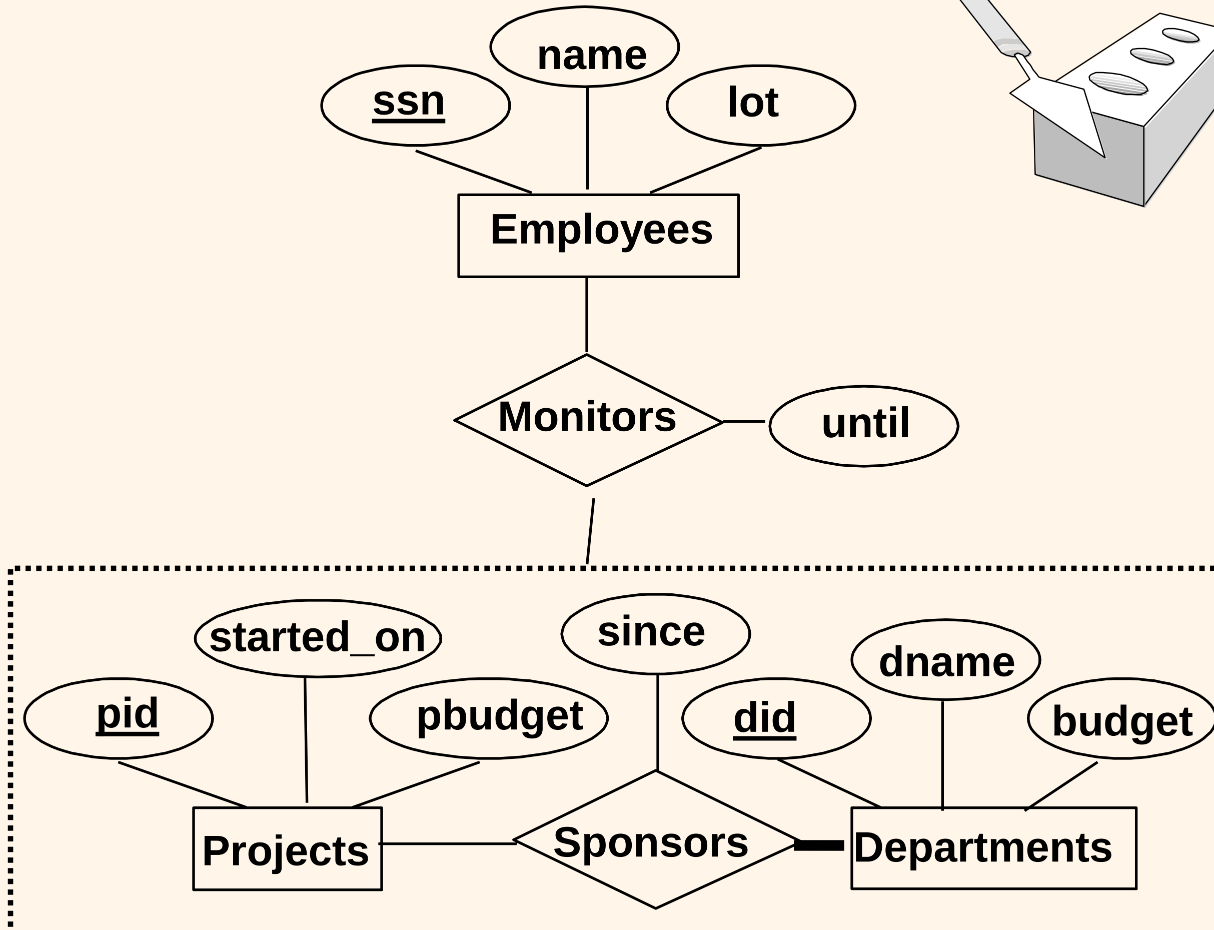
- As in C++, or other PLs, attributes are inherited.
- If we declare A **ISA** B, every A entity is also considered to be a B entity.

- *Overlap constraints*: Can Joe be an Hourly_Emps as well as a Contract_Emps entity? *(default: disallowed; A overlaps B)*
- *Covering constraints*: Does every Employees entity also have to be an Hourly_Emps or a Contract_Emps entity? *(default: no; A AND B COVER C)*
- Reasons for using ISA:
  - To add descriptive attributes specific to a subclass.
  - To identify entities that participate in a relationship.

Aggregation

❖ Used when we have to model a relationship involving (entitity sets and) a *relationship set*.

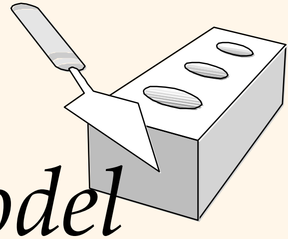
- *Aggregation* allows us to treat a relationship set as an entity set for purposes of participation in (other) relationships.



* *Aggregation vs. ternary relationship*:

❖ Monitors is a distinct relationship, with a descriptive attribute.

❖ Also, can say that each sponsorship is monitored by at most one employee.



Conceptual Design Using the ER Model

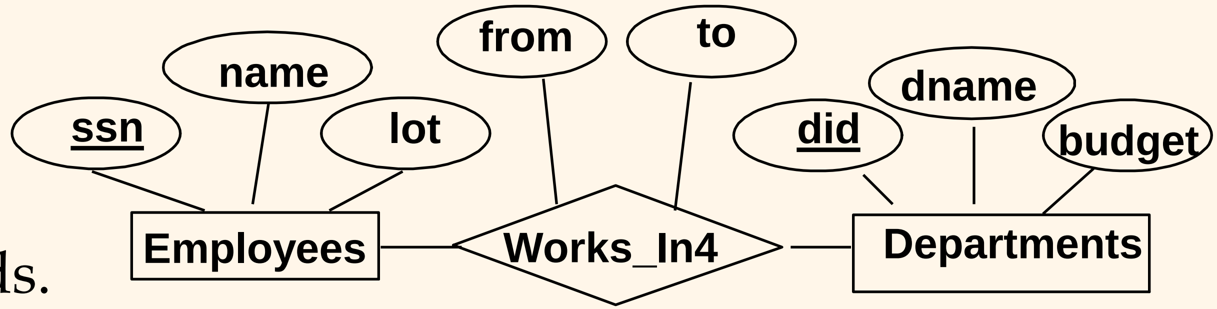
- Design choices:
  - Should a concept be modeled as an entity or an attribute?
  - Should a concept be modeled as an entity or a relationship?
  - Identifying relationships: Binary or ternary? Aggregation?
- Constraints in the ER Model:
  - A lot of data semantics can (and should) be captured.
  - But some constraints cannot be captured in ER diagrams.

Entity vs. Attribute

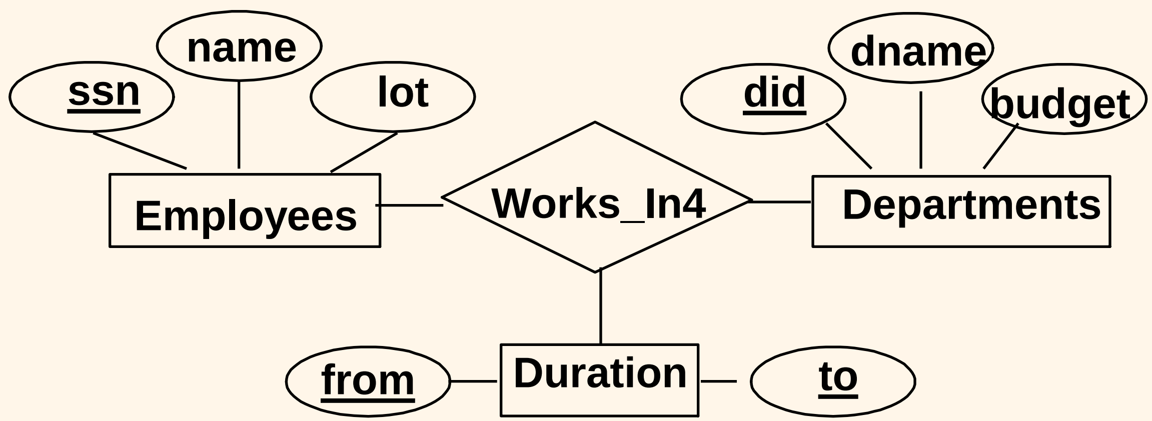
- Should *address* be an attribute of Employees or an entity (connected to Employees by a relationship)?
- Depends upon the use we want to make of address information, and the semantics of the data:
  - If we have several addresses per employee, *address* must be an entity (since attributes cannot be set-valued).
  - If the structure (city, street, etc.) is important, e.g., we want to retrieve employees in a given city, *address* must be modeled as an entity (since attribute values are atomic).

Entity vs. Attribute (Contd.)

- Works_In4 does not allow an employee to work in a department for two or more periods.

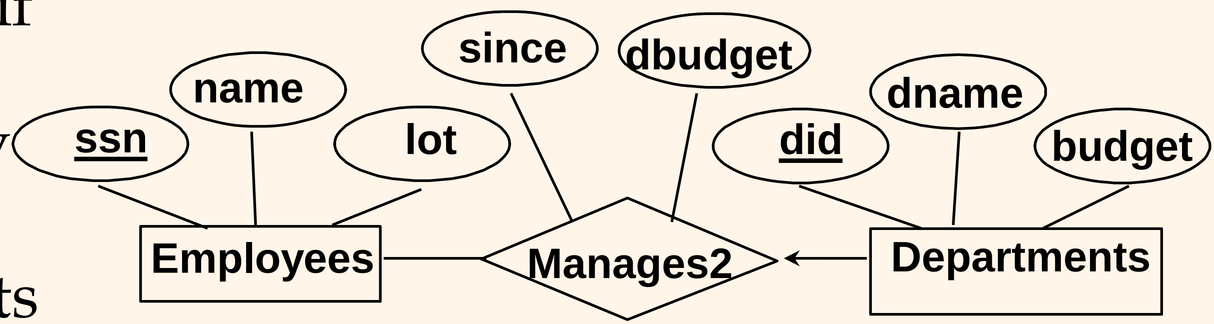


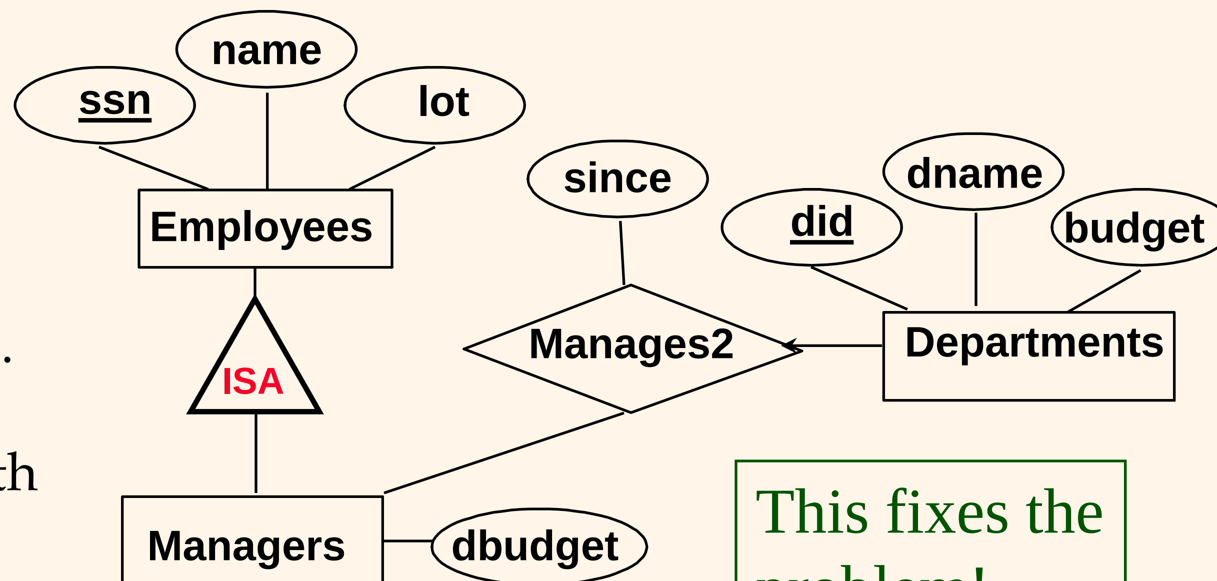
- Similar to the problem of wanting to record several addresses for an employee: We want to record *several values of the descriptive attributes for each instance of this relationship.* Accomplished by introducing new entity set, Duration.



Entity vs. Relationship

- First ER diagram OK if a manager gets a separate discretionary budget for each dept.
- What if a manager gets a discretionary budget that covers *all* managed depts?
  - Redundancy: *dbudget* stored for each dept managed by manager.
  - Misleading: Suggests *dbudget* associated with department-mgr combination.

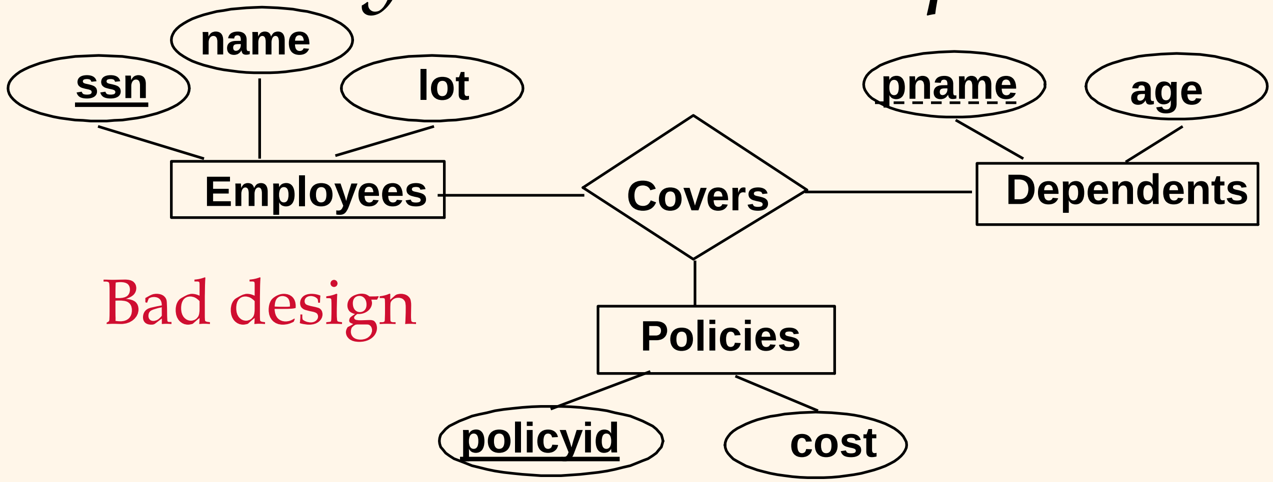




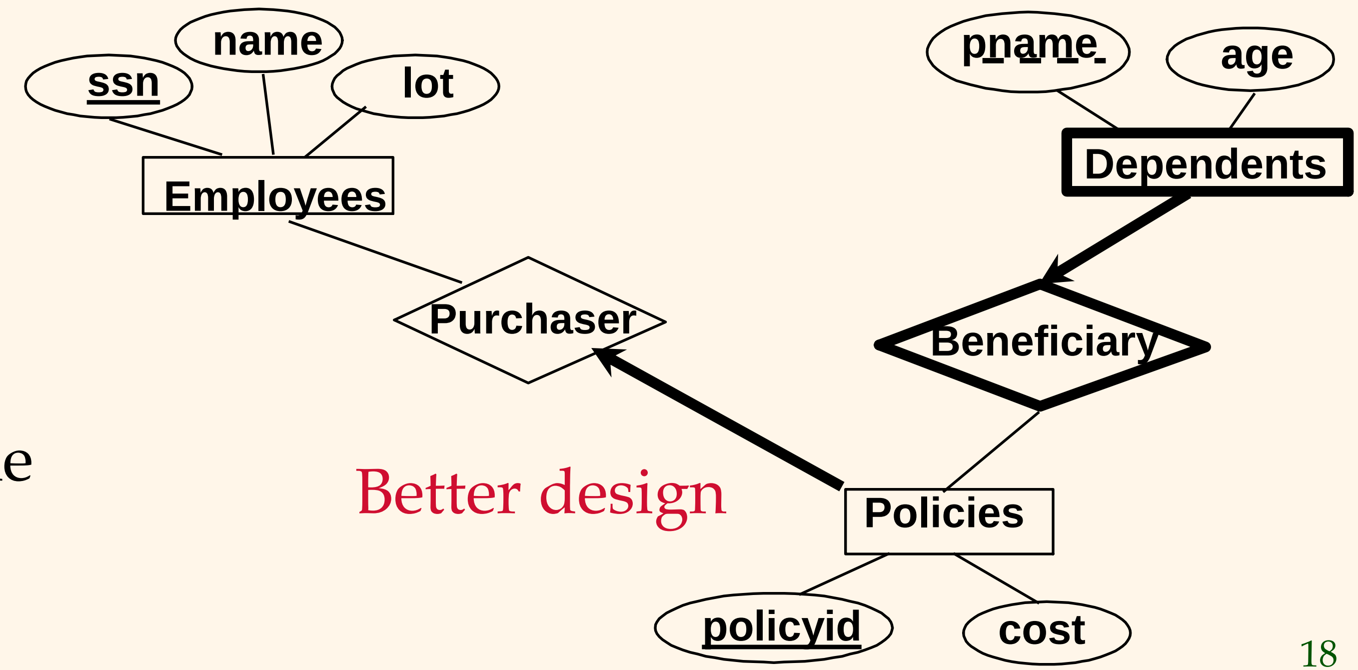
This fixes the problem!

Binary vs. Ternary Relationships

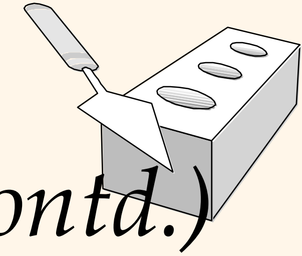
- If each policy is owned by just 1 employee, and each dependent is tied to the covering policy, first diagram is inaccurate.
- What are the additional constraints in the 2nd diagram?



Bad design



Better design

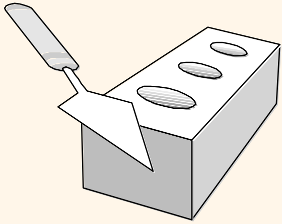


Binary vs. Ternary Relationships (Contd.)

❖ Previous example illustrated a case when two binary relationships were better than one ternary relationship.

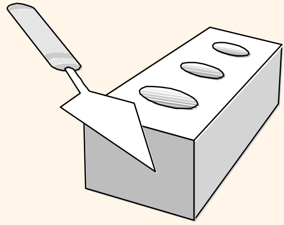
❖ An example in the other direction: a ternary relation Contracts relates entity sets Parts, Departments and Suppliers, and has descriptive attribute *qty*. No combination of binary relationships is an adequate substitute:

- S "can-supply" P, D "needs" P, and D "deals-with" S does not imply that D has agreed to buy P from S.
- How do we record *qty*?



Summary of Conceptual Design

- *Conceptual design* follows *requirements analysis*,
  - Yields a high-level description of data to be stored
- ER model popular for conceptual design
  - Constructs are expressive, close to the way people think about their applications.
- Basic constructs: *entities*, *relationships*, and *attributes* (of entities and relationships).
- Some additional constructs: *weak entities*, *ISA hierarchies*, and *aggregation*.
- Note: There are many variations on ER model.



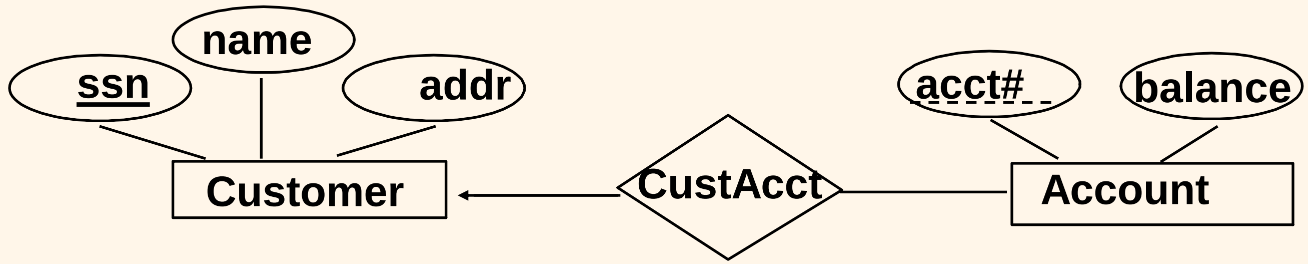
Summary of ER (Contd.)

- Several kinds of integrity constraints can be expressed in the ER model: *key constraints*, *participation constraints*, and *overlap/covering constraints* for ISA hierarchies. Some *foreign key constraints* are also implicit in the definition of a relationship set.
  - Some constraints (notably, *functional dependencies*) cannot be expressed in the ER model.
  - Constraints play an important role in determining the best database design for an enterprise.

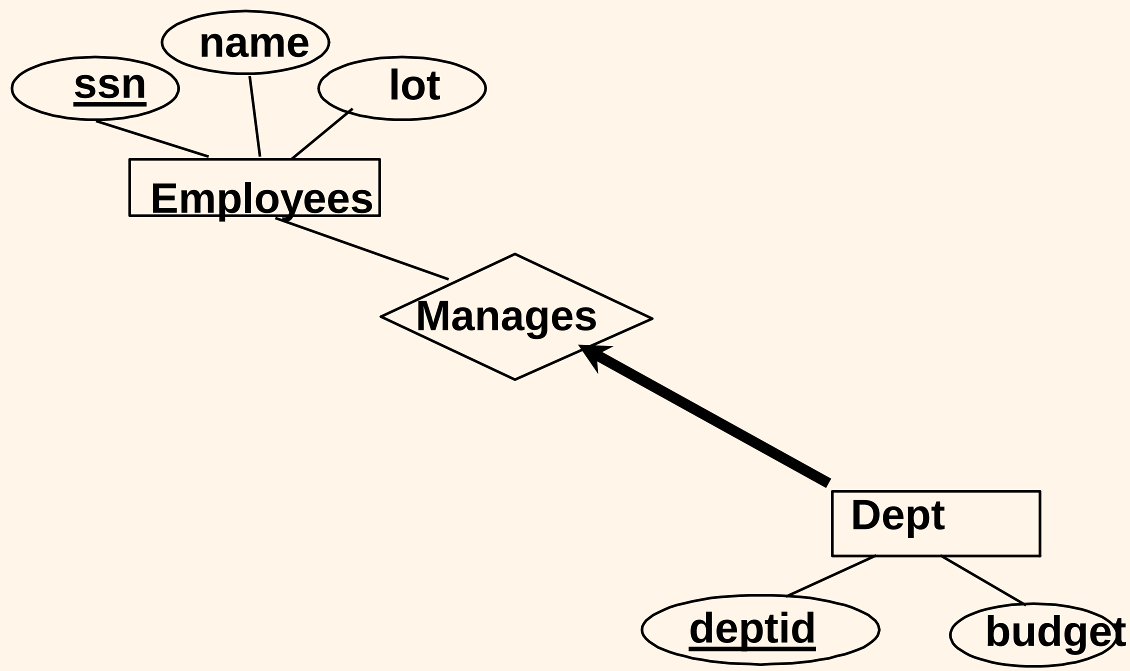
Summary of ER (Contd.)

- ER design is *subjective*. There are often many ways to model a given scenario! Analyzing alternatives can be tricky, especially for a large enterprise. Common choices include:
  - Entity vs. attribute, entity vs. relationship, binary or n-ary relationship, whether or not to use ISA hierarchies, and whether or not to use aggregation.
- Ensuring good database design: resulting relational schema should be analyzed and refined further. FD information and normalization techniques are especially useful.

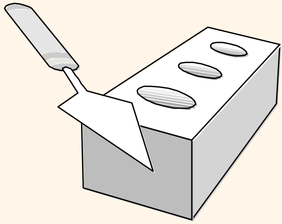
Exercise



- What can you say about policy of the bank from the ER diagram?



- What can you say about the policy of the company?



Design Exercise

- Design a DB using ER, and sketch the resulting diagram. State any important assumptions you made in reaching the design. Show explicitly whether relationships are 1-1, 1-M, or N-M.
  - UVA registrar's office: It maintains data about each class, including the instructor, students, enrollment, time and place of the class meetings. For each student-class pair, a grade is recorded.
  - Hospital: It maintains all patients visited, including age and address. It also keeps track of the information about billing, visits, data, reason for visit, and treatment.